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Next item →

1. Which of the following is NOT a way to represent a vector in two dimensions?

1 / 1 point

- ☐ Parametric form (t)
- ☐ Unit vector form (i, j)
- ☐ Cartesian coordinates (x, y)
- ☐ Polar coordinates (r, θ)

👍 Nice work

Correct. This form is not typically used to represent vectors but rather to describe curves or lines.

2. Which property of vectors states that the sum of vectors is independent of the grouping of the vectors?

1 / 1 point

- ☐ Commutativity
- ☐ Identity
- ☒ Associativity
- ☐ Distributivity

👍 Nice work

Correct. Associativity states that the grouping of the vectors does not affect their sum.

3. In the context of vector operations, what does the dot product of two perpendicular vectors equal to?

1 / 1 point

- ☐ 1
- ☐ -1
- ☒ 0
- ☐ Undefined

👍 Nice work

Correct. The dot product of two perpendicular vectors is zero.

4. Which vector property states that the dot product is distributive?

1 / 1 point

- ☐ Identity
- ☐ Commutativity
- ☒ Distributivity
- ☐ Associativity

👍 Nice work

Correct. The dot product is distributive over vector addition.

5. What is the result of the cross product of two parallel vectors?

1 / 1 point

- ☒ Zero vector
- ☐ A vector perpendicular to both
- ☐ Undefined
- ☐ A scalar

👍 Nice work

Correct. The cross product of two parallel vectors is the zero vector.

6. Which of the following properties is NOT true for vector addition?

1 / 1 point

- ☐ Commutativity
- ☐ Associativity
- ☒ Multiplicative inverse
- ☐ Identity element

👍 Nice work

Correct. Multiplicative inverse is not a property of vector addition but rather of vector multiplication by scalars.

7. What is the result of the dot product of a vector **a** with itself?

1 / 1 point

- ☒ |**a**|²
- ☐ **a**
- ☐ **a**²
- ☐ |**a**|

👍 Nice work

Correct. The dot product of a vector with itself equals the square of its magnitude.

8. If **u** = 2**i** + 3**j** and **v** = -3**i** + 2**j**, what is the magnitude of **u** + **v**?

1 / 1 point

- ☒ √26
- ☐ 1
- ☐ √29
- ☐ 0

👍 Nice work

Correct. The magnitude is calculated as √26.

9. Given the vectors **u** = 2**i** + 3**j** and **v** = -3**i** + 2**j**, calculate the vector **u** + **v**.

1 / 1 point

-1+5j

Feedback

The vector **u** + **v** = -**i** + 5**j**.

1. Add the corresponding components of the vectors:

$$\mathbf{u} + \mathbf{v} = (2\mathbf{i} + 3\mathbf{j}) + (-3\mathbf{i} + 2\mathbf{j})$$

2. Combine the i components and j components separately:

$$\mathbf{u} + \mathbf{v} = (2 - 3)\mathbf{i} + (3 + 2)\mathbf{j}$$

$$\mathbf{u} + \mathbf{v} = -\mathbf{i} + 5\mathbf{j}$$

10. If **u** = **i** - 2**j** + **k** and **v** = 2**i** + 3**j** - **k**, find the cross product **u** × **v**.

1 / 1 point

-1+3j+k

Feedback

The cross product **u** × **v** = -5**i** - 3**j** + 7**k**.

1. Write the vectors in component form:

$$\mathbf{u} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$$

$$\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$$

2. Use the cross product formula:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 2 & 3 & -1 \end{vmatrix}$$

3. Calculate the determinant:

$$\mathbf{u} \times \mathbf{v} = \mathbf{i}((-2)(-1) - (1)(3)) - \mathbf{j}((1)(-1) - (1)(2)) + \mathbf{k}((1)(3) - (-2)(2))$$

$$\mathbf{u} \times \mathbf{v} = \mathbf{i}(2 - 3) - \mathbf{j}(-1 - 2) + \mathbf{k}(3 + 4)$$

$$\mathbf{u} \times \mathbf{v} = -\mathbf{i} + 3\mathbf{j} + 7\mathbf{k}$$

11. Find the magnitude of the vector **a** = -2**i** + 3**j** + 6**k**.

1 / 1 point

7

👍 Nice work

The magnitude of **a** = 7.

1. Use the formula for the magnitude of a vector:

$$|\mathbf{a}| = \sqrt{(-2)^2 + 3^2 + 6^2}$$

2. Calculate each term inside the square root:

$$= \sqrt{4 + 9 + 36}$$

3. Sum the terms and find the square root:

$$= \sqrt{49}$$

$$= 7$$

12. Determine the angle between the vectors **u** = 4**i** - **j** and **v** = **i** + 3**j**.

1 / 1 point

85.6

Feedback

The angle between the vectors is approximately 85.60°.

1. Find the dot product:

$$\mathbf{u} \cdot \mathbf{v} = 4 \cdot 1 + (-1) \cdot 3 = 4 - 3 = 1$$

2. Calculate the magnitudes of the vectors:

$$|\mathbf{u}| = \sqrt{4^2 + (-1)^2} = \sqrt{16 + 1} = \sqrt{17}$$

$$|\mathbf{v}| = \sqrt{1^2 + 3^2} = \sqrt{1 + 9} = \sqrt{10}$$

3. Use the dot product formula to find the cosine of the angle:

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} = \frac{1}{\sqrt{17} \sqrt{10}} = \frac{1}{\sqrt{170}}$$

4. Calculate the angle:

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{170}} \right) \approx 85.60^\circ$$

13. Given the vectors **u** = **i** + 2**j** + 3**k** and **v** = 4**i** - **j** + 2**k**, calculate the dot product **u** · **v**.

1 / 1 point

8

👍 Nice work

The dot product **u** · **v** = 8.

1. Calculate the dot product:

$$\mathbf{u} \cdot \mathbf{v} = (1)(4) + (2)(-1) + (3)(2)$$

2. Compute each term:

$$= 4 - 2 + 6$$

3. Sum the terms:

$$= 8$$